

Data:
17.06.2020

Examen la disciplina ASCN :
Bilet de examen nr. 2

1.

cd \ ab	00	01	11	10
00	-	1	f	0
01	1	-	0	g
11	0	0	-	1
10	e	1	h	-

Pasul 1:

cd \ ab	00	01	11	10
00	-	1	0	0
01	1	-	0	0
11	0	0	-	1
10	e	1	0	-

$\bar{a}\bar{c}$, $\bar{a}b\bar{d}$, $\bar{a}\bar{b}c$

Pasul 2:

cd \ ab	00	01	11	10
00	-	*	f	0
01	*	-	0	g
11	0	0	-	*
10	e	*	h	-

$\bar{a}\bar{d}$, $b\bar{c}\bar{d}$, ac , $\bar{a}\bar{b}d$

Pasul 3: $\bar{a}\bar{d}\cdot e$, $b\bar{c}\bar{d}\cdot f$, $ac\cdot h$, $\bar{a}\bar{b}d\cdot g$

Pasul 4: $f^{FMD}(a,b,c,d) = \bar{a}\bar{c} + \bar{a}b\bar{d} + \bar{a}\bar{b}c + \bar{a}\bar{d}\cdot e$
 $+ b\bar{c}\bar{d}\cdot f + ac\cdot h + \bar{a}\bar{b}d\cdot g$

2

$$f_1^{FMD}(x_1, x_2, x_3) = x_1 \bar{x}_3 + x_2 x_3$$

$$f_2^{FMD}(x_1, x_2, x_3) = x_1 x_3 + \bar{x}_1 x_3 + \bar{x}_2 x_3$$

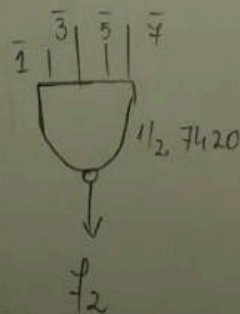
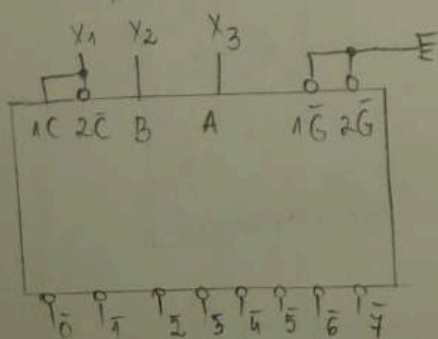
$$\begin{aligned} f_1^{FCD}(x_1, x_2, x_3) &= x_1(x_2 + \bar{x}_2) \cdot \bar{x}_3 + (x_1 + \bar{x}_1) x_2 x_3 \\ &= \underbrace{x_1 x_2 \bar{x}_3}_6 + \underbrace{x_1 \bar{x}_2 \bar{x}_3}_4 + \underbrace{x_1 x_2 x_3}_7 + \underbrace{\bar{x}_1 x_2 x_3}_3 \\ &= \sum(3, 4, 6, 7) \end{aligned}$$

$$\begin{aligned} f_2^{FCD}(x_1, x_2, x_3) &= x_1(x_2 + \bar{x}_2) x_3 + \bar{x}_1(x_2 + \bar{x}_2) x_3 + (x_1 + \bar{x}_1) \bar{x}_2 x_3 \\ &= \underbrace{x_1 x_2 x_3}_7 + \underbrace{x_1 \bar{x}_2 x_3}_5 + \underbrace{\bar{x}_1 x_2 x_3}_3 + \underbrace{\bar{x}_1 \bar{x}_2 x_3}_1 + \underbrace{x_1 \bar{x}_2 \bar{x}_3}_5 + \underbrace{\bar{x}_1 \bar{x}_2 \bar{x}_3}_1 \\ &= \sum(1, 3, 5, 7) \end{aligned}$$

a) DMUX 1:2³ si ponti logici SI-MU

$$f_1^{FCD}(x_1, x_2, x_3) = \overline{p_3 + p_4 + p_6 + p_7} = \bar{p}_3 \cdot \bar{p}_4 \cdot \bar{p}_6 \cdot \bar{p}_7$$

$$f_2^{FCD}(x_1, x_2, x_3) = \overline{p_1 + p_3 + p_5 + p_7} = \bar{p}_1 \cdot \bar{p}_3 \cdot \bar{p}_5 \cdot \bar{p}_7$$



2

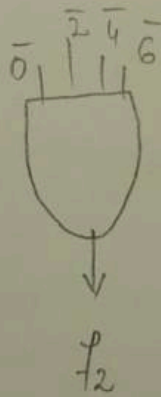
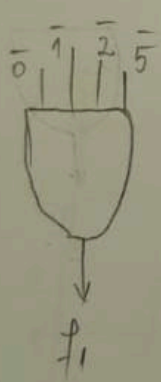
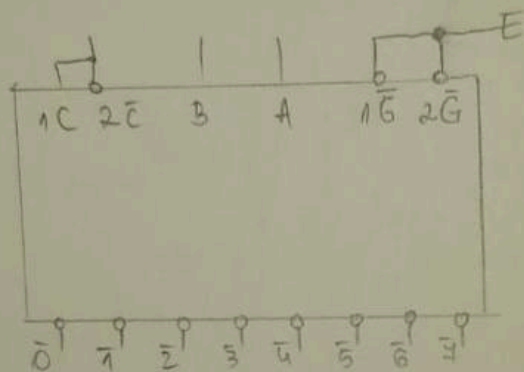
DMUX 1: 2³ si porti logice 51

$$\overline{f_1}^{FCD} = P_0 + P_1 + P_2 + P_5$$

$$f_1^{FCD} = \overline{\overline{f_1}^{FCD}} = \overline{P_0 + P_1 + P_2 + P_5} = \overline{P_0} \cdot \overline{P_1} \cdot \overline{P_2} \cdot \overline{P_5}$$

$$\overline{f_2}^{FCD} = P_0 + P_2 + P_4 + P_6$$

$$f_2^{FCD} = \overline{\overline{f_2}^{FCD}} = \overline{P_0 + P_2 + P_4 + P_6} = \overline{P_0} \cdot \overline{P_2} \cdot \overline{P_4} \cdot \overline{P_6}$$



b) MUX $2^2:1$

	x_1	x_2	x_3	f_1
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

• din prima

$$x_1 x_2 = 0, f_1 = 0 (\forall) \text{ de } x_3$$

$\Rightarrow$ pe intrare X_0 se aplică 0

• din următoarele 2 linii:

$$x_1 x_2 = 01, f_1 = 1 \text{ pt } x_3 = 1$$

$\Rightarrow$ pe intrare X_1 se aplică x_3

$$x_1 x_2 = 10, f_1 = 1 (\forall) x_3$$

$\Rightarrow$ pe X_2 se aplică 1

$$x_1 x_2 = 11, f_1 = 1 \text{ pt } x_3 = 0$$

$\Rightarrow$ pe X_3 se aplică $\bar{x}_3$

	x_1	x_2	x_3	f_2
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

$$x_1 x_2 = 00, f_2 = 1 \text{ pt } x_3 = 1$$

$\Rightarrow$ pe X_0 se aplică x_3

$$x_1 x_2 = 01, f_2 = 1 \text{ pt } x_3 = 1$$

$\Rightarrow$ pe X_1 se aplică x_3

$$x_1 x_2 = 10, f_2 = 1 \text{ pt } x_3 = 1$$

$\Rightarrow$ pe X_2 se aplică x_3

$$x_1 x_2 = 11, f_2 = 1 \text{ pt } x_3 = 1$$

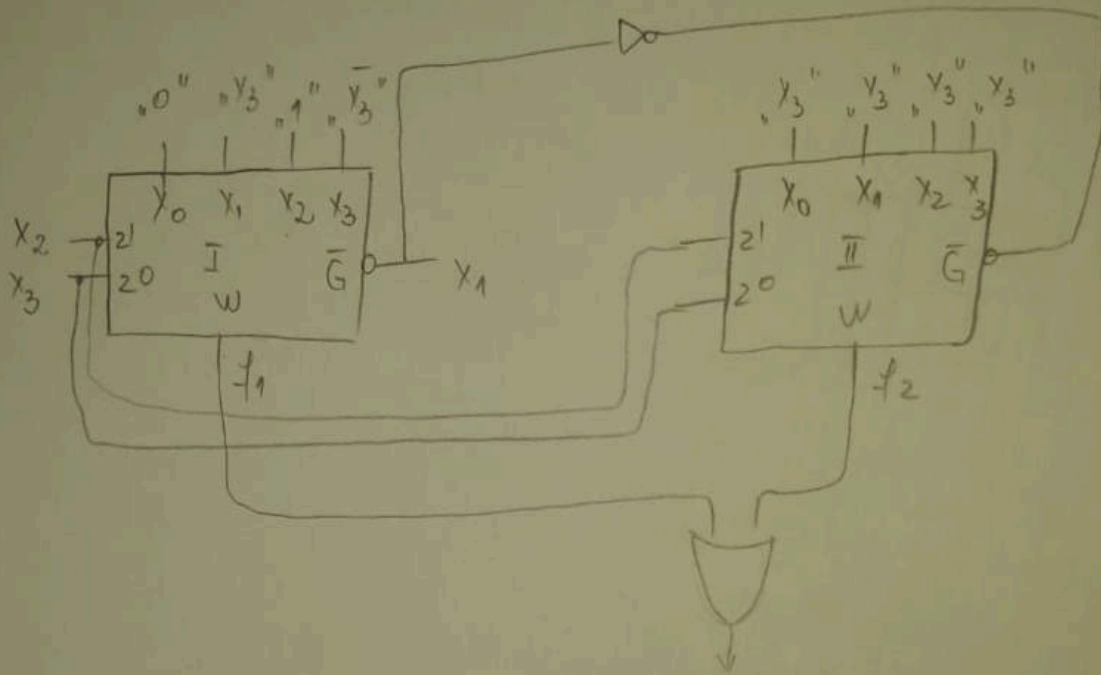
$\Rightarrow$ pe X_3 se aplică x_3

Algebraic:

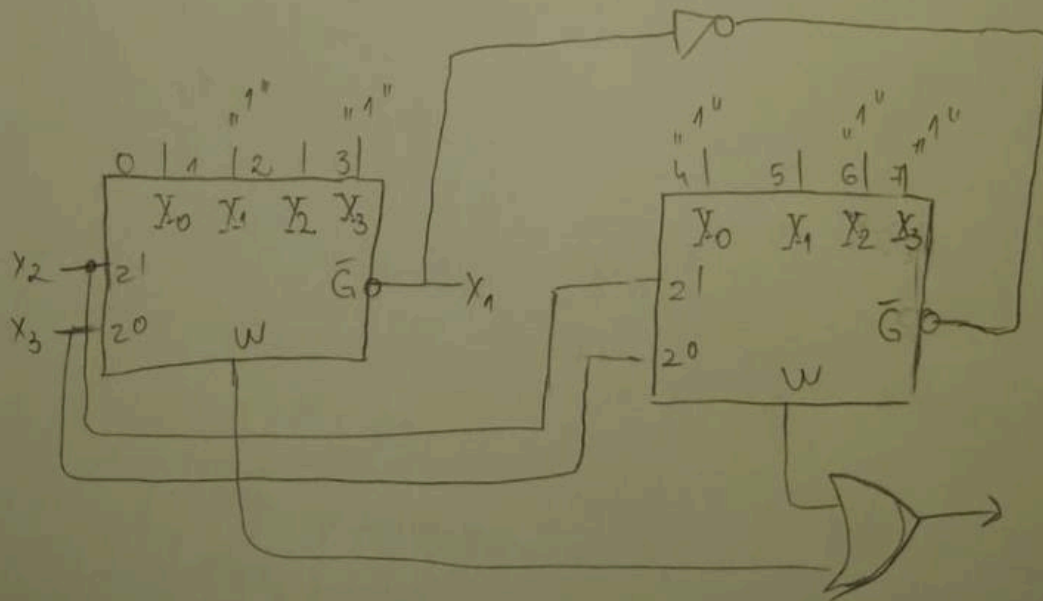
$$f_1^{FCB}(x_1, x_2, x_3) = (x_1 x_2) \cdot \bar{x}_3 + (x_1 \bar{x}_2) \bar{x}_3 + (x_1 x_2) \cdot \bar{x}_3 + (x_1 x_2) \cdot x_3 = \bar{P}_3' \cdot \bar{x}_3 + \bar{P}_2' \cdot \bar{x}_3 + \bar{P}_3' \cdot \bar{x}_3 + P_3' \cdot x_3 = \bar{P}_3' \cdot \bar{x}_3 + \bar{P}_2' \cdot \bar{x}_3 + P_3' \cdot x_3 = \bar{P}_2' \cdot \bar{x}_3 + P_3' \cdot x_3$$

$$f_{f2}^{FCD}(x_1, x_2, x_3, x_4) = (\bar{x}_1 \bar{x}_2) x_3 + (\bar{x}_1 x_2) x_3 + (x_1 \bar{x}_2) x_3 + (x_1 x_2) x_3$$

$$= \bar{P}_0' \cdot x_3 + \bar{P}_1' \cdot x_3 + \bar{P}_2' \cdot x_3 + \bar{P}_3' \cdot x_3$$



paralel :

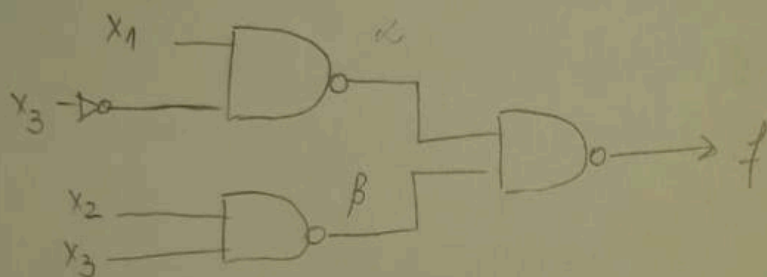


③ $f^{FMD}(x_1, x_2, x_3) = x_1 \bar{x}_3 + x_2 x_3$

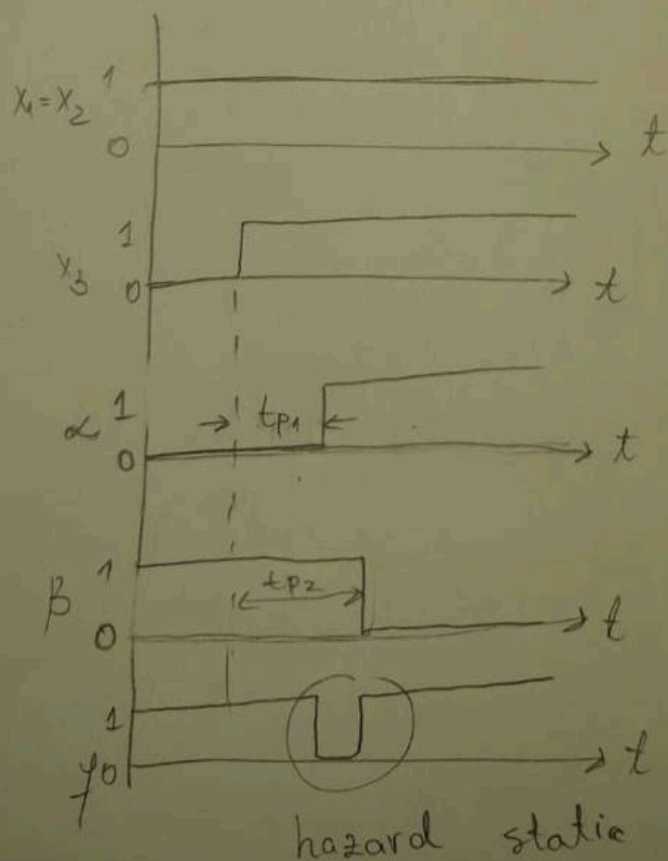
a) $x_1 \ x_2 \ x_3$
 $1 \ 1 \ 0 \rightarrow 1 \ 1 \ 1$

$x_1 x_2$				
x_3	0	1	0	1
0	0	0	1	1
1	1	1	0	0

Popica
cătălm



$x_1 = x_2 = 1$



$$\alpha = \overline{x_1 \cdot \bar{x}_3} = \begin{cases} \overline{1 \cdot 0} = 0 \\ \overline{1 \cdot 1} = 1 \end{cases}$$

$$\beta = \overline{x_2 \cdot x_3} = \begin{cases} \overline{1 \cdot 0} = 1 \\ \overline{1 \cdot 1} = 0 \end{cases}$$

$$f = \overline{\alpha \cdot \beta} = \begin{cases} \overline{1 \cdot 0} = 1 \\ \overline{0 \cdot 0} = 1 \\ \overline{1 \cdot 1} = 0 \end{cases}$$

Eliminarea hazardului:

$x_1 x_2$			
x_3	0	0	1
	1	0	1

Se adauga